41014 Integrated Project: Video Production Guide

Track A: TurtleBot3 Mobile Robot Visual Waypoint Navigation & Docking | Editor: Lachlan

CRITICAL RUBRIC RULES (KEEP STRICTLY UNDER 5:00):
1. Markers stop grading at exactly 5:00. Target run time: 4:50 to 4:55.
2. Physical hardware runs on both Track 1 and Track 2 are the required High Distinction evidence.
3. Ground truth must only appear in simulation benchmarking comparisons. Never use it in the real-time control loop.

1. Master Video Assets & Location Directory

Collect these files from Google Drive and the repository before starting your editing timeline:

Asset Name	Location	Resolution / Format	Role in Final 5-Minute Video
real_mission_camera_pov.mp4	Google Drive / Repo	640x480 @ 20 FPS (45s)	Onboard camera HUD with green ArUco box & telemetry
track_01_exterior.mp4	Google Drive	1080p / 4K Phone Video	Exterior wide shot of TurtleBot navigating Track 1
track_02_exterior.mp4	Google Drive	1080p / 4K Phone Video	Exterior shot of Track 2 proving altered obstacle robustness
terminal_screen_record.mp4	Google Drive	1080p Screen Capture	Live 20 Hz terminal telemetry stream with state estimates
mujoco_sim_run.mp4	Google Drive / Record	1080p Screen Capture	Simulation nominal obstacle avoidance and docking in 3D
part_c_ekf_performance.png	results/ in Repo	300 DPI High-Res Image	Proves 2-sigma EKF covariance contraction upon sighting
part_e_robustness_trials.png	results/ in Repo	300 DPI High-Res Image	10-trial Monte Carlo benchmark plot (100% success rate)
part_e_sim_vs_real_comparison.png	results/ in Repo	300 DPI High-Res Image	High Distinction quantitative sim vs real comparison
rowan_voiceover.m4a	Google Drive	Clean Vocal Audio	Speaker 1 voice track covering Segment 1 (0:00 - 1:30)
jonas_voiceover.m4a	Google Drive	Clean Vocal Audio	Speaker 2 voice track covering Segment 2 (1:30 - 3:00)
lachlan_voiceover.m4a	Record locally	Clean Vocal Audio	Speaker 3 voice track covering Segment 3 (3:00 - 4:55)

2. Screen Layout for Hardware Demonstration (Picture-in-Picture)

During Segment 3 (physical hardware runs), do not cut back and forth between files. Use a split screen to show all angles simultaneously:

Screen Area	Clip Source	Visual Details to Highlight
Left 60% (Main View)	Phone Exterior Recording	Full arena view (4.0m x 2.0m). Shows TurtleBot navigating smoothly past obstacles.
Top Right 40% (Inset 1)	real_mission_camera_pov.mp4	Robot POV. Features green ArUco bounding box, range r, bearing phi, and [DOCKE
Bottom Right 40% (Inset 2)	Terminal Screen Recording	Live terminal scrolling at 20 Hz. Displays EKF estimates [x, y, theta] and LiDAR clea

3. How to Record Fresh Simulation Footage (If Needed)

If you want to record the MuJoCo simulation window yourself, run these commands from the repository root:

```
cd S-C-Integrated-Project
source .venv/bin/activate
python3 main.py
```

Press the Space bar in the MuJoCo window to pause or unpause. Use your mouse to rotate the 3D camera. To reset the robot to a new random starting position, press the 'Reset' button in the right pane or run evaluate.py.

4. Word-for-Word Spoken Narration Script

Divided across Rowan, Jonas, and Lachlan. Record audio in a quiet space using phone Voice Memos at ~130 words/min.

SPEAKER 1: ROWAN (0:00 - 1:30 | 90 Seconds | 185 Words)

Visuals: Architecture diagram, state equations, coordinate frame transformation chain, split MuJoCo and real arena view.

"Welcome to our 41014 Integrated Project demonstration for Track A: Autonomous Mobile Robot Navigation and Visual Docking. Our team consists of Jonas, Lachlan, and Rowan. Our platform is a TurtleBot3 operating under unicycle kinematics, governed by state vector x, y, and theta, with forward velocity v and angular velocity omega as control inputs.

In our model analysis script, we evaluate the Lie algebra rank condition using the platform vector fields and their Lie bracket. The distribution determinant equals one across all orientations, proving small-time local controllability. For sensing, we map monocular camera pixels through intrinsic parameters into the camera frame, transform to the robot body frame, and project into the world frame.

We also evaluate the system observability matrix. Global pose is unobservable from wheel odometry alone due to dead-reckoning drift. Once the monocular camera detects the ArUco marker, the measurement Jacobian restores full rank-three observability. Crucially, our controller acts strictly on estimated states from the filter and never reads ground truth coordinates."

SPEAKER 2: JONAS (1:30 - 3:00 | 90 Seconds | 195 Words)

Visuals: MuJoCo 3D simulation running, results/part_c_ekf_performance.png, results/part_e_robustness_trials.png.

"Here is our baseline simulation running in MuJoCo. The robot starts from a randomized pose, scans the environment with a 2D LiDAR, and detects the central obstacle. To estimate state under sensor noise, we implemented an Extended Kalman Filter fusing wheel encoder odometry with camera range and bearing observations.

We tuned the process noise covariance matrix Q to reflect unmodelled wheel slip on floor surfaces over our 50-millisecond sample time, and measurement noise matrix R to match camera corner quantization. As seen in our EKF performance plot, position uncertainty bounds widen steadily while the robot bypasses the obstacle under dead reckoning.

At ten seconds, the camera sights ArUco Marker 0. The innovation covariance yields an immediate contraction in the filter covariance, pulling position error down to under three centimetres. Across our ten-trial Monte Carlo robustness benchmark with randomized starting locations, our state-feedback pipeline achieved a one hundred percent docking rate, with a mean final position error of only 3.8 centimetres and heading alignment within 1.2 degrees."

SPEAKER 3: LACHLAN (3:00 - 4:55 | 115 Seconds | 245 Words)

Visuals: Split screen Track 1 and Track 2 exterior + real_mission_camera_pov.mp4 HUD + results/part_e_sim_vs_real_comparison.png.

"To satisfy the High Distinction requirements, we deployed our complete software pipeline onto the physical TurtleBot3 in the UTS robotics laboratory using rosbridge WebSocket. Here is Track 1 on the 4.0-metre by 2.0-metre arena. On the right, the camera HUD displays our live ArUco detection box, distance r, and bearing phi.

Our LiDAR processor monitors a 100-degree frontal cone. When obstacles approach within 55 centimetres, the robot executes a reactive bypass. If an obstacle enters within 32 centimetres, the controller commands an emergency zero-velocity brake, rotating in place until the path clears. Once the marker locks, the controller transitions to Position-Based Visual Servoing and terminal corridor line-tracking, docking within three centimetres of the plate. Here is Track 2 with an altered starting condition and obstacle layout, confirming reliable docking.

In our quantitative comparison plot, Panel 1 shows that physical hardware trajectories match simulation topology closely. Panel 2 confirms that real-world 2-sigma uncertainty contracts upon visual acquisition identically to simulation.

During hardware deployment, we diagnosed two real-world issues. First, Wi-Fi transport latency caused steering oscillations under our initial visual servoing gain. We damped the gain to 0.65 and added an acceleration slew limiter to eliminate hunting. Second, we resolved the close-range waypoint singularity using terminal corridor tracking. Our pipeline proves robust, observable, and fully closed-loop docking across both simulation and real hardware. Thank you."

5. Final Video Quality Control & Submission Checklist

Run through this checklist before uploading the MP4 to YouTube and Canvas:

Check Item	Target Standard	Why It Affects Your Grade
Total Duration	Strictly 4:50 to 4:55 (Under 5:00)	Rubric Rule: Content past 5:00 is ignored. If video is 5:05, you lose the conclusion.
Audio Balancing	Normalize all 3 voices to -14 LUFS	Rowan, Jonas, and Lachlan must have matched volume without background hum o
Resolution	1080p (1920x1080) @ 30 or 60 FPS	Ensures text overlays, terminal output, and plot axes remain sharp and readable.
Ground Truth Label	Visible ONLY in Sim Benchmarks	Rubric Rule: Using ground truth in real-time control caps grade at Pass (0-49).
YouTube Visibility	Set to 'Unlisted' (NOT Private)	If set to Private, markers cannot open the link and will assign zero for video.
Title Card Info	Group Number, Member Names & IDs	Must appear in the first 5 seconds or final 5 seconds to match Canvas submission.

Submission Metadata:

- Subject: 41014 Sensors and Control in Mechatronic Systems (UTS)
- Project: Track A Mobile Robot Visual Waypoint Navigation & Docking
- Team: Rowan O'Meara, Jonas, Lachlan
- Generated PDF Guide: VIDEO_PRODUCTION_GUIDE.pdf